

Module 6: Assignment

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IFT 372: Wireless Fundamentals

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Questions

1.

a. Which security holes exist in the WEP procedures?

The main security hole existing in WEP encryption is the fact that all hosts share the same key for encrypting and decrypting transmissions and their payloads. This allowed attackers to easily use their key to intercept other devices' traffic.

b. how are they solved by WPA and WPA2 (802.1x)?

This security flaw was addressed in WPA and WPA2 by introducing temporary keys and implementing message integrity checks to ensure transmissions are not tampered with in any way.

2. How can a receiving device detect the speed at which the payload part of a frame was sent?

A receiving device can detect the speed at which the payload part of a frame was sent by examining information in its transmission headers, specifically from the Physical Layer Convergence Protocol (PLCP) header. The PLCP header has fields for things like channel coding and transfer speed.

3. What are the differences between the 'ad hoc' and 'BSS' modes of WLAN?

The difference lies in the architecture of the network. ‘ad hoc’ networks are distributed with no central hub for communication. Devices can communicate directly with others or be configured as a multi-hop network if devices need to transmit farther than one hop radio distance. Basic Service Set (BSS) networks follow more of a hub-spoke or star topology where all traffic passes through a central access point.

4. What disadvantages does the DCF method have for telephony and video streaming applications?

DCF works by choosing a random interval delay for retransmitting data when it senses that the channel is busy. This results in a variable delay interval which conflicts with the constant delay interval needed by services like telephony and video streaming.

5. What is an ESS?

An Extended Service Set (ESS) is similar to a BSS but differs in the fact that it uses multiple access points with the same SSID (madhurihammad, 2023). Devices connected to the network can switch between the access points depending on signal strength. This effectively provides more extensive coverage for users.

6. What is an SSID and in which frames is it used?

The Service Set ID is a string that identifies a WLAN. It is constantly broadcast in the beacon frame so that users can see available networks and connect to one of their choosing with the

appropriate credentials. The SSID is then saved on their device so that they can reconnect automatically in the future.

7. What is the maximum transfer rate that can be reached in a data transfer between two 802.11g devices in a BSS?

The theoretical top transfer speed between two 802.11g devices in a BSS is about 54 Mbps. The actual speed is much lower than that, but the network's nature and the implementation of certain mechanisms make the actual transfer much slower. For example, control packets that limit flow such as ACK packets reduce speeds by a large factor. Also, in a BSS, traffic must go through the central access point, which adds an intermediate hop to each transmission between two devices.

8. What kinds of PS mechanisms exist in the WLAN standard?

Devices can enter Power Saving Mode when there is no traffic in order to conserve power for more purpose-related processes. A device will send an empty frame with a power save bit set to 1 to indicate that it is entering power saving mode and disables its transceiver. The access point then starts to queue traffic destined for the device. The device periodically checks the Traffic Indication Map (TIM) in the beacon frame. If there is traffic, the the device will exit power saving mode and handle the traffic. If not, it will continue to sleep.

9. Which additional functionalities can often be found in WLAN APs?

WLAN APs often times have extra Ethernet ports for connecting wired devices to bridge them with the network. APs also often come with a DHCP server running to auto-configure devices with an IP address, subnet mask, DNS servers, default gateway, etc.

10. Why are ACK frames used in a WLAN?

ACK frames are used in WLAN because the wireless medium is much less reliable than wired mediums. Wireless transmission is very prone to errors so ACK frames are used to make sure the traffic gets to its destination in the way they were transmitted and were not lost or distorted due to things like electromagnetic interference.

11. Why are three MAC addresses required in BSS frame?

The three MAC addresses correspond to the physical interfaces involved in the WLAN transmission. They are the source MAC address, access point mac address, and the destination MAC address. The AP MAC address is required because traffic must be relayed through the AP and the sending and receiving devices know where to transmit to.

12. Why do 802.11g networks use the RTS/CTS mechanism?

802.11g networks use the RTS/CTS mechanism to accommodate for older standards. WLANs following the 802.11g standard use a different modulation and coding schemes as 802.11b which means 802.11b devices would not register the new signals as a valid transmission and see the channel as free. RTS/CTS allows for this backwards compatability with older devices that may not use 802.11g.

References

madhurihammad. (2023, May 12). *Difference between BSS and ESS*. GeeksforGeeks.

<https://www.geeksforgeeks.org/difference-between-bss-and-ess/>